Epistemic Uncertainty for Test-Time Discovery

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Abstract

Automated scientific discovery with large language models requires finding genuinely new solutions, yet test-time training systems plateau: average reward improves but the maximum stops rising. We show this is caused by diversity collapse. The policy concentrates on familiar patterns because standard reinforcement learning penalizes uncertain mutations. A single model cannot fix this, because it cannot tell whether its uncertainty comes from a hard problem or from regions it has simply never learned. We introduce UG-TTT, which adds a small ensemble of low-rank adapters over a frozen base model. Token-level mutual information measures where the model is unsure, and this signal is added as an exploration bonus in the policy update. A nuclear norm regularizer keeps the adapter versions different from each other, preserving the exploration signal throughout training. On four scientific discovery benchmarks, UG-TTT improves maximum reward on three tasks, sustains substantially higher solution diversity, and ablations confirm the regularizer is essential for maintaining exploration.

1 Introduction

Large language models (LLMs) are increasingly deployed not merely as assistants but as active participants in scientific discovery, proposing hypotheses, generating candidate algorithms, and searching vast combinatorial spaces that were previously accessible only through expert intuition Qi et al. [2023], Yuksekgonul et al. [2026a], Rankovic et al. [2025]. Systems such as AlphaEvolve and TTT-DISCOVER have shown that an LLM coupled with a verification loop can rediscover and occasionally improve on human-designed solutions for open mathematical problems. Yet the same systems reveal a persistent ceiling: they improve average performance while the maximum reward, which is the quantity discovery actually requires, plateaus early Yuksekgonul et al. [2026a], Thomas et al. [2025b]. Understanding why this plateau is structural rather than incidental is the starting point of our work.

Discovery is an out-of-distribution task. A genuine discovery is a solution that lies beyond prior human knowledge and, consequently, outside any pretrained model's training distribution. This framing has an immediate consequence: any method that searches high-probability regions of a model's learned distribution primarily performs retrieval or exploitation. However, scientific discovery demands ideas beyond the training data. Therefore, progress hinges on a key question: how can we push an LLM to navigate its knowledge boundaries, where its initial predictive confidence is low, but the potential for meaningful epistemic shift is high?

Reinforcement learning as a reward. TTT-DISCOVER runs RL on a single problem: after PUCT selects a parent solution, the policy generates mutations that a verifier scores. Since discovery requires a single breakthrough solution, what matters is whether the policy ever generates something genuinely novel. The standard reinforcement learning (RL) objective maximizes expected reward, which rewards reliability and penalizes variance Cheng et al. [2025], Li et al. [2025]. Under this objective, the policy gradient points toward mutations whose outcomes are already well-estimated,

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UG-TTT: Uncertainty Guided TTT

$$\mathcal{L}(\Theta) \;=\; \sum_{t=1}^{|o|} \min\Big[\, \frac{\pi_\Theta(o_t|q,o_{<t})}{\pi_{\Theta_{\text{old}}}(o_t|q,o_{<t})}\,(A_t^{\text{ent}} + \gamma\, R_{\text{MI}}),\; \text{clip}\left(\frac{\pi_\Theta(o_t|q,o_{<t})}{\pi_{\Theta_{\text{old}}}(o_t|q,o_{<t})},\, 1-\epsilon,\, 1+\epsilon\right)(A_t^{\text{ent}} + \gamma\, R_{\text{MI}})\Big] \;-\; \beta\, \|U_\Theta\|_*$$

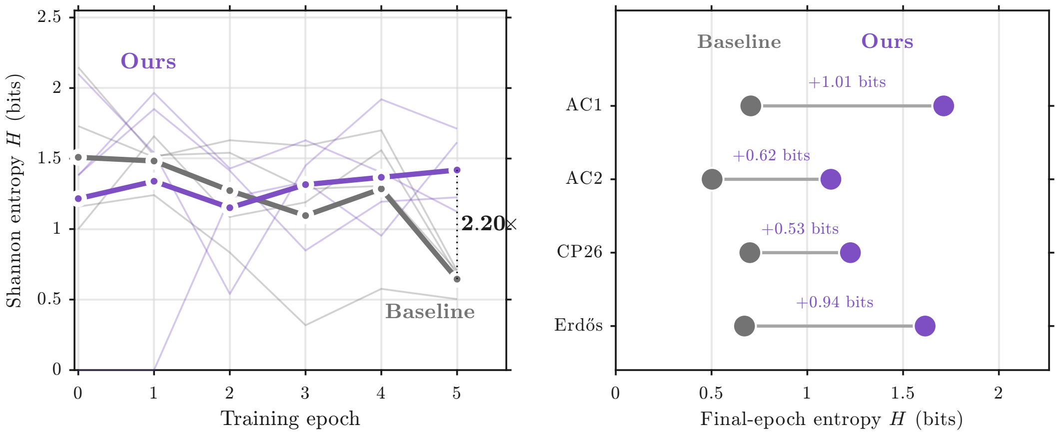


Figure 1: **UG-TTT preserves the diversity that drives discovery.** The training objective (top) augments the TTT-DISCOVER clipped importance-sampling (IS) surrogate with a mutual-information (MI)-shaped advantage and a nuclear-norm regulariser (purple terms). *Left:* Per-epoch Shannon entropy H of the algorithmic-template distribution on AC1 across five LoRA adapters; UG-TTT (purple) sustains $H \approx 1.4$ bits while the TTT-DISCOVER (gray) collapses to 0.65 bits by epoch 5, a $2.20\times$ gap. *Right:* Final-epoch entropy gains: $+1.01$ bits (AC1), $+0.62$ (AC2), $+0.53$ (CP26), $+0.94$ (Erdős), directly enabling the $R_{\max}$ improvements of Sec. 4.

because uncertain mutations contribute high-variance gradients that the optimizer suppresses. The result, observed empirically across test-time training systems, is diversity collapse: average reward rises while the maximum plateaus Yuksekgonul et al. [2026a], Thomas et al. [2025b].

Why existing fixes are insufficient. Recent work has recognized this pathology and introduced signals that encourage exploration, such as entropy bonuses, confidence-based difficulty weighting, and uncertainty-guided reasoning budgets Cheng et al. [2025], Li et al. [2025]. All of these measure uncertainty from a single trained network, which fundamentally cannot distinguish two very different sources. *Aleatoric uncertainty* reflects ambiguity in the input, such as a question phrased multiple ways, and is irreducible. *Epistemic uncertainty* reflects the model's own ignorance about regions it has not adequately learned, and is exactly what we want to target Balabanov and Linander [2024], Abbasi-Yadkori et al. [2024]. A single model produces high entropy in both cases and cannot tell them apart. For discovery this is a fatal confound: the method reliably spends compute exploring ambiguous prompts while the true knowledge frontier, where breakthroughs live, receives no special signal. A concurrent MCTS-based approach, AutoDiscovery [**?**], illustrates both problems at once: it uses Bayesian surprise derived from sampling a single frozen LLM, inheriting the aleatoric-epistemic confound, and because model weights are never updated, it has no mechanism to internalize domain knowledge as discoveries accumulate.

Disagreement as a principled signal. Bayesian theory decomposes total predictive uncertainty into aleatoric (irreducible data noise) and epistemic (model ignorance) components Hüllermeier and Waegeman [2019], Shorinwa et al. [2024]. While an ensemble's total predictive entropy captures both, mutual information between predictions and model parameters specifically isolates the epistemic component as disagreement among plausible hypotheses over the weights Abbasi-Yadkori et al. [2024]. When multiple independently trained copies of a model agree, even on a broad high-entropy distribution, the region exhibits only aleatoric uncertainty: the model knows it is uncertain, but all members agree on that uncertainty. When they disagree, the region is genuinely uncertain in the epistemic sense. Ensembles thus expose a signal no single model can. Full LLM ensembles are computationally infeasible, but Low-Rank Adaptation (LoRA) ensembles approximate the same signal at a fraction of the cost: several lightweight low-rank adapters share a frozen base model and each acts as an independent posterior sample Wang et al. [2023], Balabanov and Linander [2024]. Disagreement among these adapters flags inputs the model is uncertain about in the specific sense that matters for discovery, namely that its knowledge there is incomplete. Such an ensemble, however, only functions as a compass if it does not collapse. Jointly trained adapters sharing data and objective have no mechanism preventing their weight configurations from converging, and when they do the disagreement signal drops to numerical zero and all navigational value disappears Zhai et al. [2023a]. Our measurements confirm this: an unregularized five-adapter ensemble reaches mutual information below 10^{-5} within two epochs. A diversity constraint on the ensemble parameters is therefore a prerequisite, not an optimization detail.

Contributions. We introduce Uncertainty-Guided Test-Time Training UG-TTT, which augments the reinforcement learning objective of test-time discovery Yuksekgonul et al. [2026b] with a Bayesian epistemic uncertainty signal derived from a low-rank adapter ensemble over a shared frozen base. UG-TTT maintains K LoRA adapters jointly updated under the same policy objective and treats their disagreement, quantified as the token-level mutual information (MI) $\mathrm{MI}_t = H(\bar{p}(\cdot \mid y_{<t})) - \frac{1}{K}\sum_{k=1}^{K} H(p_k(\cdot \mid y_{<t}))$, as a posterior estimate of how far a generation lies from the region the model has adequately learned. This signal is standardized within each rollout group and injected into the policy advantage through a gain coupled to the entropic temperature β, yielding a second optimization axis that rewards epistemic surprise proportionally to, and only conditional on, comparable execution quality. To prevent the degeneracy in which jointly trained adapters converge to a common weight configuration and the mutual information collapses to numerical zero, we introduce a nuclear-norm regularizer on the stacked adapter matrices that enforces subspace diversity across ensemble members and preserves the uncertainty signal throughout training. We choose this regularizer because maximizing the nuclear norm of the stacked adapter matrices carries a clear geometric guarantee: it drives each LoRA ensemble member into a mutually orthogonal input subspace, directly counteracting subspace collapse and empirically preserving a usable epistemic uncertainty signal throughout RL training, whereas prior LoRA regularizers focus on redundancy reduction without this explicit link to uncertainty preservation. The combined construction produces an adaptive curriculum in which exploration pressure scales with β: it remains inactive while the policy is still acquiring basic competence and intensifies precisely when execution rewards plateau, the regime in which standard reinforcement learning provably collapses to low-entropy exploitative behavior. Empirically, UG-TTT sustains 200–650× higher ensemble mutual information than an unregularized baseline, retains 1.2–1.8 bits of solution-family entropy where the baseline converges to a single family, and improves the maximum reward attained on three benchmarks from the TTT-DISCOVER suite at strictly lower token cost (Fig. 1).

2 Background and Related Work

RLHF and reward model optimization. Reinforcement learning from human feedback (RLHF) has become the standard mechanism for aligning LLMs with human preferences Havrilla et al. [2024], Hong et al. [2025], Rafailov et al. [2024], producing gains in reasoning, commonsense, and code generation through Proximal Policy Optimization (PPO)-style optimization against a learned reward model DeepSeek-AI et al. [2025], Havrilla et al. [2024], Hong et al. [2025]. Its central pathology is reward hacking: scalar rewards compress nuanced objectives into a single dimension, enabling exploitation of proxy loopholes and inducing overconfidence, hallucination, and distributional fragility Khalaf et al. [2025], Rafailov et al. [2024], Xia et al. [2025], Shorinwa et al. [2024], Farquhar et al. [2024], Tao et al. [2025]. The structural limitation relevant to us is that outcome-based scalar rewards inherently suppress exploration.

Test-time training. Test-time training (TTT) adapts model parameters at inference using self-supervised signals or retrieved neighbors Hu et al. [2025], Hardt and Sun [2023], Akyürek et al. [2024], often with parameter-efficient adapters to mitigate forgetting Hu et al. [2025], Moradi et al. [2025], and has delivered substantial gains on the Abstraction and Reasoning Corpus (ARC) and BIG-Bench Hard Akyürek et al. [2024]. Like RLHF, it inherits mode collapse under sparse outcome rewards Thomas et al. [2025a]: without dense intrinsic signals, adaptation converges prematurely on familiar solution paths rather than advancing toward the knowledge frontier.

Uncertainty quantification. Foundational work on uncertainty quantification (UQ) separates aleatoric from epistemic uncertainty and proposes Bayesian networks, deep ensembles, Monte Carlo dropout, and evidential methods as estimation strategies Abdar et al. [2020], Charpentier et al.

[2022], Hüllermeier and Waegeman [2019], Caldeira and Nord [2020]. In LLMs, UQ has been deployed for hallucination detection through token probabilities, verbalized confidence, ensemble disagreement, and conformal intervals Xia et al. [2025], Shorinwa et al. [2024], Farquhar et al. [2024], Tao et al. [2025], Liu et al. [2025], Vashurin et al. [2024], Huang et al. [2023], with production systems such as LM-Polygraph Fadeeva et al. [2023] and entropy-based confabulation detectors Farquhar et al. [2024]. These methods remain post-hoc: they flag unreliable outputs but do not integrate into the training signal, and most fail to cleanly separate epistemic from aleatoric sources at LLM scale Liu et al. [2025], Abdar et al. [2020].

Intrinsic motivation and MI-based exploration. Curiosity bonuses and entropy regularization have been proposed to counter exploitation bias in RL with LLMs Du et al. [2023], Liao et al. [2025], Gao et al. [2025], Dwaracherla et al. [2024], Levy et al. [2025], Cheng et al. [2025], though classical novelty rewards suffer vanishing incentives in large action spaces Dai et al. [2025], Yuan et al. [2022]. Recent work uses ensemble disagreement, including LoRA-based variants, as an intrinsic signal tied to the knowledge boundary Shorinwa et al. [2024], Hüllermeier and Waegeman [2019], with actor-wise perplexity bonuses and critic-wise value variance integrated into advantage functions Dai et al. [2025]. Frameworks such as Curiosity-Driven Exploration (CDE) and i-MENTOR demonstrate that maintained entropy prevents premature convergence in sparse-reward reinforcement learning with verifiable rewards (RLVR) Dai et al. [2025], Gao et al. [2025], but balancing intrinsic against outcome rewards risks calibration collapse Dai et al. [2025]. None of these approaches target epistemic uncertainty at the granularity required for frontier exploration, which is the gap UG-TTT addresses.

3 Uncertainty-Guided Test-Time Training

Optimising R_{exec} alone supplies no signal to distinguish a solution the policy already assigns high probability from one at the genuine edge of its knowledge. The core intuition behind UG-TTT is simple: if several independently adapted copies of the model disagree about what token comes next, that position is one the model has not yet resolved, which is exactly where a discovery-oriented policy should focus its search. UG-TTT operationalises this intuition with three coupled additions to the TTT-DISCOVER training loop. First, a K-member LoRA ensemble over a shared frozen base produces per-token mutual information MI_t, a Bayesian disagreement measure that localises positions where the model is genuinely uncertain rather than merely facing a hard input. Second, a rollout-level summary U_i derived from MI_t is injected into the policy-gradient advantage as a shaped bonus A_i^{shaped}, scaled automatically to the current reward magnitude and the entropic temperature so exploration pressure rises precisely when the policy begins to converge. Third, a nuclear-norm regulariser $\mathcal{L}_{\text{NNM}}$ on the stacked adapter down-projections prevents the ensemble from collapsing to shared weights and silencing the MI_t signal, because disagreement-based exploration is only as useful as the diversity it measures. Section 3.1 recalls the TTT-DISCOVER setup that UG-TTT inherits unchanged; Sections 3.2–3.4 define each component in turn; Section 3.5 assembles them into the per-step objective $\mathcal{L}_{\text{total}}$ and shows that TTT-DISCOVER is recovered when $\lambda_{\text{NNM}} = 0$.

3.1 Setup

An instance of AI-driven scientific discovery is an environment $\mathcal{E} = (d, \mathcal{S}, \mathcal{A}, R, \mathcal{T})$ with natural-language task description d, solution space $\mathcal{S}$, action set $\mathcal{A}$ of next-token continuations, verifiable scalar reward $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ returned by a deterministic program checker, and transition $\mathcal{T}$ that appends an action $a \in \mathcal{A}$ to a parent state $s \in \mathcal{S}$. A policy $\pi_\theta(a \mid q, s)$ generates a response $o = (o_1, \ldots, o_{|o|})$ of thinking tokens and executable code, conditioned on a prompt q that embeds d and s. The discovery objective is the per-instance maximum reward $\max_{(s,a)} R(s, a)$ rather than its expectation: a single breakthrough solution dominates the metric, and a method that improves average reward while leaving the maximum unchanged has not advanced discovery.

We adopt the TTT-DISCOVER training setup without modification: at each step, G rollouts per parent state are drawn via a PUCT-curated history $\mathcal{H}$, scored with $R_{\text{exec}}^{(i)}$, and assigned leave-one-out (LOO) entropic advantages A_i through an adaptive temperature $\beta(s)$ that maintains a fixed KL budget against uniform. The full objective, temperature bisection, and clipped importance-sampling loss $\mathcal{L}_{\text{PG}}$ are stated in Appendix 7.

3.2 Ensemble posterior and token-level epistemic uncertainty

Optimizing R_{exec} alone, even under the entropic objective of Sec. 3.1, leaves the policy without any signal that distinguishes a confident continuation from one the model has barely learned to score: the gradient pulls toward whichever solution family the base model already prefers, and the maximum-reward metric that discovery cares about plateaus early (Sec. 1). UG-TTT addresses this by replacing the single adapter of TTT-DISCOVER with an ensemble whose per-token predictive disagreement supplies a Bayesian epistemic-uncertainty signal. Disagreement localizes the knowledge frontier: positions where the posterior over weights spreads across incompatible next-token hypotheses, so rewarding high-disagreement rollouts pulls the policy toward generations it has not yet learned to be confident about.

Ensemble of low-rank adapters. For every tracked linear layer we maintain K low-rank perturbations on top of the shared frozen weights,

$$\Theta^{(k)} \;=\; \Theta_{\text{base}} + B^{(k)}A^{(k)}, \qquad A^{(k)} \in \mathbb{R}^{r\times d_{\text{in}}},\; B^{(k)} \in \mathbb{R}^{d_{\text{out}}\times r},\; r \ll \min(d_{\text{in}}, d_{\text{out}}), \tag{1}$$

for $k \in \{1, \dots, K\}$, where Θ_{base} are the frozen pretrained weights of the layer, $A^{(k)}$ and $B^{(k)}$ are the trainable down- and up-projection matrices of adapter k, and r is the shared LoRA rank. Rollouts are drawn one adapter at a time, round-robin (we choose G as a multiple of K, so each adapter produces G/K rollouts per group); scoring evaluates the complete sequence through all K adapters in a single chunked forward pass and returns the full-vocabulary next-token distributions $p_{\theta_k}(\cdot \mid q, o_{<t})$ at every position t. Round-robin governs only generation: at training time every adapter receives the gradient from all G rollouts of the group (Sec. 3.5, Algorithm 1). The K adapters act as samples from a variational posterior $q(\theta \mid \mathcal{D})$ whose predictive is the arithmetic mixture $\bar{p}_t(\cdot) = \frac{1}{K}\sum_{k=1}^{K} p_{\theta_k}(\cdot \mid q, o_{<t})$.

Token-level mutual information. With each adapter supplying its own per-token predictive $p_{\theta_k}(\cdot \mid q, o_{<t})$, the mixture $\bar{p}_t$ pools all K views into an approximate posterior predictive. This pooling makes disagreement measurable: positions where every member agrees contribute little to the spread of $\bar{p}_t$, while positions where members diverge across incompatible next tokens mark the knowledge frontier the policy has not yet resolved. The per-position mutual information quantifies this disagreement as the standard Bayesian Active Learning by Disagreement (BALD) decomposition,

$$\text{MI}_t \;=\; H(\bar{p}_t) \;-\; \frac{1}{K}\sum_{k=1}^{K} H\big(p_{\theta_k}(\cdot \mid q, o_{<t})\big)\,, \tag{2}$$

where H is Shannon entropy over the vocabulary V. Eq. (2) is the standard BALD decomposition [Houlsby et al., 2011, Cao and Tsang, 2021, He et al., 2026]: it is non-negative by Jensen's inequality, vanishes exactly when all members agree, and subtracts the aleatoric (in-member) entropy so only the epistemic component remains. Both terms are evaluated on the full (K, chunk, V) logit tensor before it is released; a chunked implementation that keeps peak memory bounded is described in Appendix 10.

Top-7% aggregation. MI_t is defined at each token position, but the policy-gradient update operates on full rollouts, not individual tokens. Moreover, most positions in a code rollout carry near-zero MI_t: structural tokens such as indentation, closing brackets, and boilerplate keywords produce negligible disagreement across adapters. Averaging uniformly would dilute the pivotal decision points, where algorithm choices, loop bounds, or key constants create genuine disagreement, with uninformative filler. We therefore compress the per-token signal into a single rollout-level uncertainty score by retaining only the most uncertain positions,

$$U_i \;=\; \underset{t\in[1,|o_i|]}{\text{top-7\%-mean}}\, \text{MI}_t^{(i)}. \tag{3}$$

We adopt 7% as a heuristic order-of-magnitude choice, consistent with evidence that a small minority of decoding steps carry the pivotal RL signal [Wang et al., 2025, AdaDec authors, 2025]; the value adapts to sequence length so an n-token response contributes $\lceil 0.07n \rceil$ positions. Calibration details are in Appendix 10.

3.3 Uncertainty-shaped advantage

U_i from Eq. (3) gives us a per-rollout measure of how uncertain the ensemble is about that rollout's content. The goal now is to turn this scalar into a policy-gradient bonus, but two obstacles stand in the way. First, raw U varies by orders of magnitude across tasks, so a fixed coefficient that works on one benchmark over-explores or under-explores on another. Second, exploration pressure should intensify precisely when the policy begins to converge toward a mode, not run at a constant rate that competes with reward-seeking regardless of training stage. Two properties therefore make U_i usable as a policy-gradient signal: it must be standardized within each rollout group so the coefficient transfers across tasks; and it must be coupled to the entropic temperature $\beta(s)$ so exploration pressure rises automatically as the policy sharpens. The shaping below realizes both, while preserving the reward-seeking direction of A_{exec} exactly. We compute A_{exec} and $\beta(s)$ from R_{exec} alone, so the adaptive temperature schedule that drives TTT-DISCOVER's max-seeking behavior is unaffected by the exploration signal. We then center and clip U within each group,

$$\tilde{U}_i \;=\; \mathrm{clip}\!\left(\frac{U_i - \overline{U}}{\sigma_U},\; -3,\; 3\right), \qquad \overline{U} = \tfrac{1}{G}\sum_i U_i,\;\; \sigma_U = \sqrt{\tfrac{1}{G-1}\textstyle\sum_i (U_i - \overline{U})^2}, \tag{4}$$

where σ_U is the unbiased sample standard deviation; this yields a zero-mean, bounded signal commensurable across tasks. The clip at ± 3 handles fat-tailed outliers from isolated pathological positions. The exploration coefficient is coupled to $\beta(s)$,

$$\gamma_{\text{eff}}(s) \;=\; \alpha \cdot \min\!\left(\frac{\beta(s)}{\beta_{\text{ref}}},\; \gamma_{\max}\right), \tag{5}$$

with base rate $\alpha = 0.1$, reference $\beta_{\text{ref}} = 2.0$, and safety clip $\gamma_{\max} = 10$. The shaped advantage is

$$A_i^{\text{shaped}} \;=\; A_i \;+\; \gamma_{\text{eff}}(s) \cdot \overline{|A|} \cdot \tilde{U}_i\,, \qquad \overline{|A|} \;=\; \frac{1}{G}\sum_j |A_j|. \tag{6}$$

The prefactor $\overline{|A|}$ rescales the MI bonus to the current magnitude of A_{exec}: in a near-uniform group where $|A|$ is small the bonus is proportionally damped, while in a decisive group it sits on the same order as A_{exec} and genuinely competes. The per-adapter loss $\mathcal{L}_{\text{PG}}^{(k)}(\theta_k)$ and its KL-adjusted form are stated in Appendix 7.

Proposition 1 (*Bounded scale perturbation of the shaped advantage*). *Let $z_i = (U_i - \overline{U})/\sigma_U$ and $\mathcal{C} = \{i : |z_i| > 3\}$. By Eq.* (4)*, $\sum_{i=1}^{G} \tilde{U}_i = 0$ when $\mathcal{C} = \varnothing$ (and is treated as 0 when $\sigma_U = 0$). In general,*

$$\left|\sum_{i=1}^{G} \tilde{U}_i\right| \;\le\; \sum_{i\in\mathcal{C}} \big(|z_i| - 3\big) \;\le\; |\mathcal{C}|\big(\sqrt{G-1} - 3\big),$$

where the last bound uses $\sum_i z_i^2 = G - 1$ from the unbiased σ_U in Eq. (4)*. Consequently*

$$\left|\sum_i A_i^{\text{shaped}} - \sum_i A_i\right| \;\le\; \gamma_{\text{eff}}(s)\,\overline{|A|}\,|\mathcal{C}|\big(\sqrt{G-1} - 3\big),$$

which vanishes whenever the clip is inactive ($\mathcal{C} = \varnothing$) and is otherwise driven entirely by the clipped excess $|z_i| - 3$ on $\mathcal{C}$. The bonus is intentionally biased in the policy-gradient sense: the directional bias toward high-MI rollouts is what drives exploration, in line with the broader advantage-shaping perspective of 6. Proposition 1 bounds the residual interaction between this bias and the magnitude of the reward gradient by the clipped excess on $\mathcal{C}$; clipping is uncommon in practice because U_i seldom concentrates more than three standard deviations away from the group mean for groups of the size we use ($G \le 64$). Proof in Appendix 9.

Remark 1 (*β-coupling aligns exploration with collapse onset*). Under the entropic objective, $\beta(s)$ grows monotonically as the policy sharpens around a mode: $\beta \to \infty$ is precisely the regime in which TTT-DISCOVER collapses. The coupling in Eq. (5) ramps γ_{eff} up at exactly the rate at which reward-seeking pressure tightens, so exploration intensifies in lock-step with convergence without any hand-tuned schedule.

3.4 Subspace diversity regulariser

The shaped advantage of Sec. 3.3 amplifies exploration wherever the ensemble disagrees, but this immediately raises the question: what prevents the ensemble from converging to a common solution? Nothing in the shared RL objective does. When all K adapters receive the same rollout gradients under the same reward signal, they are pushed toward the same optima, and disagreement is merely an incidental byproduct of random initialization that erodes over time. In pilot runs the K adapters collapsed within two to three epochs: pairwise row-space cosines approached 1 and U_i fell below 10^{-3} (Table 3), at which point MI_t dropped to zero and the entire exploration mechanism built in the preceding sections silently degraded into standard RL on a single adapter.

Construction. We prevent collapse by maximizing the nuclear norm of the per-layer stacked down-projections (Nuclear-Norm Maximisation, NNM), which forces each adapter to occupy a distinct rank-r input subspace. For each tracked linear layer ℓ, let

$$W_\ell \;=\; \big[A_\ell^{(1)};\, A_\ell^{(2)};\, \dots;\, A_\ell^{(K)}\big] \;\in\; \mathbb{R}^{Kr\times d_{\mathrm{in}}}, \qquad \mathcal{L}_{\mathrm{NNM}} \;=\; -\frac{1}{L}\sum_{\ell=1}^{L} \big\|W_\ell\big\|_*, \tag{7}$$

[1] where $A_\ell^{(k)}$ is the down-projection of adapter k at layer ℓ, $\|\cdot\|_*$ is the nuclear norm (the sum of singular values), and L is the number of tracked layers. We minimize $\mathcal{L}_{\mathrm{NNM}}$ jointly with the RL loss; the negative sign converts the minimization into a maximization of the aggregate nuclear norm.

Proposition 2 (*The global maximum is subspace-orthogonal*). *Fix a common adapter-wise Frobenius norm $\|A_\ell^{(k)}\|_F = c$ for all $k \in \{1,\dots,K\}$. Whenever $Kr \le d_{\mathrm{in}}$, the global maximum of $\|W_\ell\|_*$ on this constraint set consists of K adapter blocks $\{A_\ell^{(k)}\}_{k=1}^{K}$ whose row spaces are mutually orthogonal in $\mathbb{R}^{d_{\mathrm{in}}}$, and each block has r equal singular values, all equal to $c/\sqrt{r}$.* Proof in Appendix 9.

Proposition 2 makes the geometric intent of $\mathcal{L}_{\mathrm{NNM}}$ precise: maximizers of the nuclear norm are configurations in which each adapter attends to a distinct rank-r input subspace, which is exactly the condition under which the BALD decomposition in Eq. (2) yields a non-trivial signal. The equal-Frobenius-norm assumption is invoked only to expose the geometry; $\mathcal{L}_{\mathrm{NNM}}$ alone does not enforce it, and in practice scale equalization across adapters arises implicitly from AdamW and shared LoRA initialization. We regularize only the down-projection $A^{(k)}$ and leave the up-projection $B^{(k)}$ unconstrained, matching the empirical finding of Zhai et al. [2023b] that joint regularization of both projections induces oscillatory co-adaptation: $A^{(k)}$ determines which input directions an adapter attends to, while $B^{(k)}$ remains free to produce whatever output magnitude the RL signal demands. A Frobenius-normalized, scale-invariant variant is described in Appendix 10; all experiments in this paper use Eq. (7).

3.5 Training objective and a degenerate special case

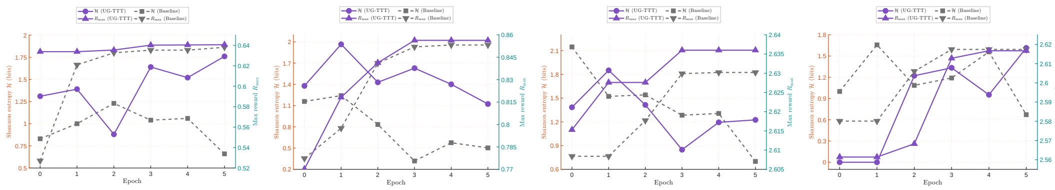
The three components introduced above, the ensemble policy-gradient loss of Sec. 3.3, the frozen-base KL anchor inherited from Yuksekgonul et al. [2026b], and the diversity regularizer of Eq. (7), combine into a single per-step objective

$$\mathcal{L}_{\mathrm{total}} \;=\; \underbrace{\frac{1}{K}\sum_{k=1}^{K}\mathcal{L}_{\mathrm{PG}}^{(k)}(\theta_k)}_{\text{ensemble RL}} \;+\; \underbrace{\lambda_{\mathrm{KL}}\,\mathrm{KL}\big(\pi_\theta \,\|\, \pi_{\mathrm{base}}\big)}_{\text{base anchor}} \;+\; \underbrace{\lambda_{\mathrm{NNM}}\,\mathcal{L}_{\mathrm{NNM}}}_{\text{diversity}}. \tag{8}$$

Defaults are $\alpha = 0.1$, $\beta_{\mathrm{ref}} = 2.0$, $\gamma_{\max} = 10$, $\lambda_{\mathrm{KL}} = 0.01$, $\lambda_{\mathrm{NNM}} \in \{0.05, 0.10\}$, and $K = 5$. The KL anchor uses the per-token reward-shaping estimator of Yuksekgonul et al. [2026b] (Appendix 7); the full training loop is Algorithm 1 (Appendix 8).

Proposition 3 (TTT-DISCOVER *as a degenerate special case of* UG-TTT). *Fix $\lambda_{\mathrm{NNM}} = 0$ together with either (a) $K = 1$, or (b) $K > 1$ with all members initialized to identical weights. Then $U_i \equiv 0$, $\tilde{U}_i \equiv 0$, the shaped advantage A_i^{shaped} reduces to A_i, and the diversity term vanishes*

[1] Throughout, $A_\ell^{(k)}$ denotes a LoRA down-projection matrix at layer ℓ of adapter k, while A_i denotes the scalar leave-one-out rollout advantage of Sec. 3.1 (Appendix 7, Eq. (11)); the two are distinguished by their index style and by context.



(a) AC1 (b) AC2 (c) CP26 (d) Erdős

Figure 2: **Training dynamics on the four UG-TTT benchmarks.** Each panel reports, across six training epochs on a single problem, the Shannon entropy of the algorithmic family distribution over correct rollouts (left axis, orange) together with the cumulative maximum reward $R_{\max}$ (right axis, teal). Dashed curves correspond to the baseline and solid curves to UG-TTT. The baseline exhibits monotone entropy contraction on every problem, whereas UG-TTT sustains an elevated entropy throughout training while concurrently matching or surpassing the reward ceiling of the baseline. The phenomenon is invariant across the four problem geometries, indicating that the preservation of ensemble diversity is concomitant with, rather than antagonistic to, the optimisation of $R_{\max}$.

from the objective. Under these substitutions, Eq. (8) *coincides with the* TTT-DISCOVER *objective of Yuksekgonul et al. [2026b] and the* UG-TTT *update step-for-step reproduces the baseline update. Branch (b) is well-posed because each adapter computes* $\mathcal{L}_{\mathrm{PG}}^{(k)}$ *over all* G *rollouts of the group (not only the* G/K *it generated), so identically-initialized members receive identical gradients and remain tied throughout training.*

Proposition 3 isolates the contribution of UG-TTT precisely. The epistemic-uncertainty signal in Eq. (2), the shaped advantage in Eq. (6), and the subspace regularizer in Eq. (7) are the only terms that distinguish $\mathcal{L}_{\mathrm{total}}$ from the TTT-DISCOVER objective, and every gain reported in Sec. 4 is attributable to their joint effect.

4 Experiments

We evaluate UG-TTT on four problems from the TTT-DISCOVER benchmark suite [Yuksekgonul et al., 2026b]: a step-function autocorrelation inequality (**AC1**, 1000-element sequence, minimise the bound C_1); the autocorrelation *lower*-bound problem (**AC2**, maximise C_2); circle packing of n=26 unit-square circles (**CP26**, maximise sum of radii); and the Erdős minimum-overlap integral (**Erdős**, minimise C_5). Three questions organise the section: (i) does UG-TTT discover better solutions across problems (Sec. 4.1); (ii) is preserved ensemble diversity the mechanism behind those gains (Sec. 4.2, 4.3); and (iii) which components are load-bearing (Sec. 4.4).

Setup. All runs use Qwen3-8B with K=5 LoRA adapters of rank 16, NNM coefficient λ_{NNM}=0.075, MI advantage gain α=0.1, and G=64 rollouts per epoch for 6 epochs on a single RTX Pro 6000 (96 GB), using the TTT-DISCOVER prompts and reward verifiers unchanged.

4.1 Main results

Table 1 reports the maximum reward $R_{\max}$ reached during training and the Shannon entropy of the algorithmic-template distribution over correct rollouts at the final epoch. UG-TTT strictly improves $R_{\max}$ on three of the four problems and ties on the fourth, while keeping 1.1–1.7 bits of late-training template entropy on every problem (Fig. 2). The baselines collapse to H<0.71 bits in all four cases.

The numerical Δs are small in absolute terms; their meaning lies in where they sit on the discovery curve. On AC1 UG-TTT crosses the $R \geq 0.63$ threshold at *step 12* (epoch 0); the baseline first reaches it at step 189 (epoch 2), a 16× delay, and never crosses 0.64 within the same compute budget while UG-TTT reaches it at step 219. On CP26 the +0.0058 gain takes UG-TTT into a regime that the publicly reported TTT-DISCOVER system reaches only with 50 training steps and a much larger model (gpt-oss-120B), *matching* their 2.635983 result with our 6-epoch / Qwen3-8B configuration. On AC2 the +0.0031 ceiling flip is produced by the `differential_evolution` family which yields *zero* correct rollouts under the unregularised ensemble (Table 2). The Erdős row is a tie within 10^{-3} on the scalar bound but the *how* differs: UG-TTT records 13 new-best events

Table 1: **Main results across four UG-TTT benchmarks.** On every problem UG-TTT either improves $R_{\max}$ or matches it; in all four cases it preserves 1.1–1.7 bits of template entropy where the baseline collapses below 0.71 bits. $R_{\max}$: higher is better on every problem (rewards converted to $\uparrow$ form per the TTT-DISCOVER convention). The CP26 UG-TTT ceiling matches the published TTT-DISCOVER state of the art (2.6360 vs. 2.635983 [Yuksekgonul et al., 2026b]).

	$R_{\max}$ during training			H at epoch 5 (bits)	
Problem	Baseline	UG-TTT	Δ	Baseline	UG-TTT
AC1	0.6381	**0.6406**	+0.0025	0.70	**1.71**
AC2	0.8532	**0.8563**	+0.0031	0.50	**1.12**
CP26	2.6302	**2.6360**	+0.0058	0.70	**1.23**
Erdős	**2.6174**	2.6167	−0.0007	0.67	**1.62**

Streaming MI: enabled on AC1 and CP26; disabled on AC2 and Erdős (paired calibration in §4.4, full diagnostic in Appendix 11).

Table 2: **Method-exclusive discoveries (idea emergence).** For each problem, the algorithmic motif used by the highest-reward rollout, its adoption among the 30 best rollouts, and its adoption among all correct rollouts during training. Top-30: number of the 30 highest-reward rollouts that contain the motif. Adoption: motif uses divided by total correct rollouts. The baseline never crosses 1/30 on any problem.

		Top-30 (count)		Adoption (%)	
Problem	Winning template	Baseline	UG-TTT	Baseline	UG-TTT
AC1	`target_sum` $= \sqrt{2n}$ (Cauchy–Schwarz)	1/30	**18**/30	3.0	**28.0**
AC2	`scipy diff_evo` $\cup$ gradient-based	0/30	**13**/30	0.9	**21.6**
CP26	non-uniform rows, e.g. `[5,5,5,5,6]`	1/30	**13**/30	17.6	**22.4**
Erdős	Fourier init $h(x) = a + b\sin(kx)$	0/30	**6**/30	0	**6.9**

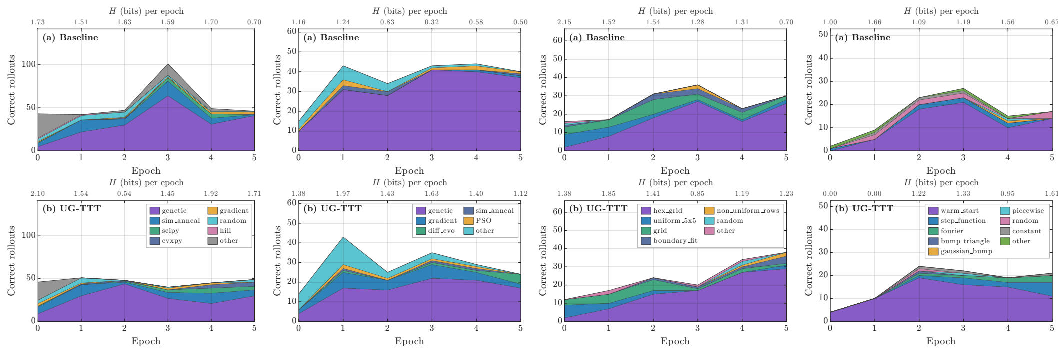
versus 7 for the baseline (+86%) and nearly doubles the discretisation resolution (mean $n_{\text{points}} = 170$ vs. 89), surfacing a method-exclusive Fourier-basis initialisation; we return to this in Sec. 4.4.

4.2 Diversity is the mechanism

Figure 3 traces, for every problem, the per-epoch mixture of algorithmic families used by all correct rollouts. The story is identical across the four problems and across two distinct kinds of "family" (outer search loop on AC1/AC2; structural initialisation on CP26/Erdős): the baseline starts wide, narrows monotonically, and finishes at $H \le 0.71$ bits with one family taking 82–93% of correct rollouts. UG-TTT either holds a wider portfolio throughout (AC1, CP26) or starts tight and *opens up* as training progresses (AC2, Erdős), finishing every problem with ≥ 3 live families. The collapse of the baseline is not a quirk of one problem: it is a *universal failure mode* of TTT-DISCOVER-style RL on discovery problems, and the diversity regulariser is what prevents it. This preserved entropy is a direct consequence of the MI bonus. As β rises and reward-seeking pressure tightens, γ_{eff} scales proportionally (Remark 1, Eq. (5)), sustaining exploration pressure on underrepresented families at precisely the point where the baseline contracts. The ablation in Sec. 4.4 confirms this mechanistically: without NNM the MI signal collapses to $\sim 10^{-5}$ within two epochs and the family distribution narrows identically to the single-adapter baseline.

4.3 Method-exclusive discoveries

Preserving entropy is only useful if the surviving families produce *better* programs. The surviving families translate into winning motifs the baseline reaches at most 1/30 times on any problem (Table 2); two are mathematically substantive—the Cauchy–Schwarz target $\sqrt{2n}$ on AC1 (a closed-form symmetry bounding C_1 from below) and the Fourier ansatz $h(x) = a + b\sin(kx)$ on Erdős—and both lie inside families the baseline extinguishes by the final epoch (Fig. 3). Preserved ensemble diversity is the precondition for entering them at all.



(a) AC1 (b) AC2 (c) CP26 (d) Erdős

Figure 3: **Diversity collapse is universal and is averted by UG-TTT.** Per-epoch composition of correct rollouts by algorithmic family on each of the four TTT-DISCOVER benchmarks, arranged as a 1×4 row. Within each panel, the *top row* is the baseline (1 LoRA, no MI/NNM) and the *bottom row* is UG-TTT (5 LoRA + MI + NNM); columns share a y-scale so the two conditions are directly comparable, and the strip at the top reports the Shannon entropy of the family distribution at the first and last epoch ($H_{\text{start}} \to H_{\text{end}}$). Family vocabularies are problem dependent and identical between the two conditions, so the contrast lies in which families remain populated rather than which are sampled. At the terminal epoch $H = 1.71/1.12/1.23/1.62$ bits under UG-TTT versus $H = 0.70/0.50/0.70/0.67$ bits under the baseline, evidencing a uniform contraction of the baseline portfolio that is absent under UG-TTT.

Table 3: **CP26 component ablation.** Removing NNM collapses ensemble MI within two epochs; the MI bonus then becomes identically zero and the run reverts to an unregularised 5-LoRA RL baseline. Mean MI is per-rollout, averaged over epoch 5. "Tokens to own $R_{\max}$" is cumulative `rollout/num_tokens` until the run first reaches its own ceiling; "Tokens to $R = 2.6302$" is the cost of *matching the no-NNM ceiling*—a direct compute comparison at fixed reward.

Configuration	$R_{\max}$	Mean MI (ep. 5)	Tokens to own $R_{\max}$	Tokens to R=2.6302
Full UG-TTT (NNM + MI bonus)	**2.6360**	$\mathbf{2.2 \times 10^{-2}}$	$\mathbf{2.33 \times 10^{6}}$	$\mathbf{2.04 \times 10^{6}}$
No NNM (MI bonus collapses to 0)	2.6302	6.6×10^{-6}	2.72×10^{6}	2.72×10^{6}

4.4 Ablation and computational cost

Table 3 compares the full method against the MI-without-NNM configuration on CP26 (the same 5-LoRA run used as the comparator in Table 1). Without NNM the ensemble's row spaces align within two epochs, ensemble MI falls to $\sim 10^{-5}$, the MI bonus becomes identically zero, and the run is indistinguishable from unregularised 5-LoRA RL: $R_{\max}$ stalls at 2.6302 and the full method matches that ceiling on 25% fewer training tokens (2.04×10^{6} vs. 2.72×10^{6}). NNM is the load-bearing component: it keeps the ensemble informative so that the MI bonus has a non-trivial signal to convert into exploration pressure on the policy gradient.

Streaming MI is enabled by default, with paired calibration on AC2 and Erdős. Early stopping in streaming mode (see Appendix 11) caps the thinking phase and re-checks ensemble MI before continuing generation. This is the default during training. On AC1 and CP26, it triggers in 50.0% and 55.6% of rollouts respectively (saving an average of 569 and 757 tokens per rollout; $5\times$ the unregularised CP26 baseline at 10.7%). These savings drive the 25% token-budget reduction reported in Table 3. The Table 1 $R_{\max}$ rollouts on AC1 and CP26 also used streaming, so streaming-on is the configuration for these results.

On AC2 and Erdős, streaming did not outperform the baseline, so we used paired no-streaming controls. Streaming reduced $R_{\max}$ by 0.0148 on AC2 ($0.8563 \to 0.8415$; mean rollout length 8,711 $\to$ 5,312 tokens) and by 0.0017 on Erdős ($2.6167 \to 2.6150$; 8,720 $\to$ 5,255 tokens). The reduction on AC2 is meaningful; on Erdős, it is within variance. Thus, streaming is disabled on AC2 and Erdős, and we report the no-streaming results in Table 1. No length–quality diagnostic predicts this: rank correlation between thinking-phase length and execution reward (epochs 0–2) is not significantly positive (Appendix 11), so calibration is empirical. Paired no-streaming control on AC1/CP26 is addressed in the multi-seed follow-up (Sec. 2). The Erdős tie in Table 1 shows the method matches the bound but finds a distinct family (Fourier basis with SLSQP at $n_{\text{points}} = 400$) not found by the single-adapter baseline.

5 Conclusion

We argue that diversity collapse in test-time discovery systems is a structural consequence of single-model, scalar-reward optimization: policy gradients suppress high-variance candidates, and a single network cannot disentangle epistemic from aleatoric uncertainty. UG-TTT addresses both by introducing an ensemble of low-rank adapters that provides a token-level mutual information signal, combined with a nuclear norm regularizer to maintain subspace diversity and a temperature-coupled exploration bonus.

UG-TTT improves maximum reward on three of four benchmarks and matches the fourth while discovering qualitatively distinct solutions, sustaining substantially higher mutual information and late-stage entropy. Ablations identify the nuclear norm regularizer as critical, with its removal causing rapid collapse of the uncertainty signal and reversion to standard reinforcement learning behavior.

6 Appendix

7 Baseline objective: TTT-DISCOVER in full notation

We collect here, for reference, the TTT-DISCOVER training signal that UG-TTT builds on. TTT-DISCOVER optimises a state-conditional entropic objective with an adaptive temperature $\beta(s)$,

$$J_\beta(\theta) \;=\; \mathbb{E}_{s\sim\text{reuse}(\mathcal{H})}\left[\frac{1}{\beta(s)}\log\,\mathbb{E}_{a\sim\pi_\theta(\cdot|q,s)}\Big[e^{\beta(s)\,R_{\text{exec}}(s,a)}\Big]\right], \tag{9}$$

which interpolates between expected reward ($\beta \to 0$) and the per-state maximum ($\beta \to \infty$). The outer expectation is over parent states s drawn from the reuse distribution over the PUCT history $\mathcal{H}$, and the inner expectation is over rollouts a sampled from the policy at prompt q and parent s. For every group, $\beta(s)$ is selected as the unique $\beta > 0$ such that the softmax $q_\beta(a) \propto \exp(\beta R_{\text{exec}}(a))$ has a fixed $\ln 2$ KL budget against uniform,

$$\mathrm{KL}\big(q_\beta \,\big\|\, \mathrm{Unif}\big) \;=\; \ln 2, \tag{10}$$

which exists and is unique whenever the within-group rewards are not all identical, since $\mathrm{KL}(q_\beta \| \mathrm{Unif})$ is continuous and strictly increasing in β on $(0, \infty)$ with limits 0 and $\ln G$. We solve Eq. (10) by 60 iterations of bisection on $[0, 10^6]$; on the degenerate equal-reward set, the bisection saturates at the upper bracket and we fall back to $A_i \equiv 0$. Given $\beta(s)$, rollout $i \in \{1, \dots, G\}$ receives the leave-one-out entropic weight and advantage

$$w_i \;=\; \frac{\exp\big(\beta\, R_{\text{exec}}^{(i)}\big)}{\frac{1}{G-1}\sum_{j\neq i}\exp\big(\beta\, R_{\text{exec}}^{(j)}\big)}, \qquad A_i \;=\; w_i - 1, \tag{11}$$

which centres the advantage within each group without a learned value function. The rollouts are trained with the clipped importance-sampling loss

$$\mathcal{L}_{\text{PG}}(\theta) \;=\; -\sum_{t=1}^{|o|}\min\Big[\rho_t(\theta)\,A_i,\; \text{clip}\big(\rho_t(\theta),\, 1{-}\epsilon,\, 1{+}\epsilon\big)\,A_i\Big], \tag{12}$$

with likelihood ratio $\rho_t(\theta) = \pi_\theta(o_t \mid q, o_{<t}) / \pi_{\theta_{\text{old}}}(o_t \mid q, o_{<t})$, reducing to plain importance sampling when ϵ is disabled. A KL-to-base penalty $\lambda_{\text{KL}}\,\mathrm{KL}(\pi_\theta \| \pi_{\text{base}})$ is added when $\lambda_{\text{KL}} > 0$.

The per-adapter form used inside UG-TTT replaces A_i with A_i^{shaped} (Eq. (6)):

$$\mathcal{L}_{\text{PG}}^{(k)}(\theta_k) \;=\; -\sum_{t=1}^{|o|}\min\Big[\rho_t^{(k)}(\theta_k)\,A_i^{\text{shaped}},\; \text{clip}\big(\rho_t^{(k)},\, 1{-}\epsilon,\, 1{+}\epsilon\big)\,A_i^{\text{shaped}}\Big], \tag{13}$$

with per-adapter ratio $\rho_t^{(k)}(\theta_k) = \pi_{\theta_k}(o_t \mid q, o_{<t}) / \pi_{\theta_{k,\text{old}}}(o_t \mid q, o_{<t})$.

Token-wise KL adjustment. When $\lambda_{\text{KL}} > 0$, the scalar A_i^{shaped} is broadcast to a per-token tensor and the token-wise correction of Yuksekgonul et al. [2026b] is applied before the clipped-IS reduction. Concretely, for each position t the advantage entry is adjusted by $-\lambda_{\text{KL}}\left(\log \pi_\theta(o_t \mid q, o_{<t}) - \log \pi_{\text{base}}(o_t \mid q, o_{<t})\right)$, which under sampling from π_θ gives the standard k1 single-sample stochastic estimator of $\nabla_\theta\, \lambda_{\text{KL}}\text{KL}(\pi_\theta \| \pi_{\text{base}})$ folded into the per-token PG target—no second autograd pass through the policy is needed. The transcription is identical to the reference implementation and is stated here only to fix notation used in the proof of Proposition 1.

8 Full training algorithm

9 Proofs

9.1 Proof of Proposition 1 (bounded scale perturbation of the shaped advantage)

Fix a parent state s and its group of G rollouts. If $\sigma_U = 0$, Eq. (4) sets $\tilde{U}_i \equiv 0$ by convention, and $A_i^{\text{shaped}} = A_i$ identically. Otherwise

$$\sum_{i=1}^{G} \tilde{U}_i \;=\; \sum_{i=1}^{G} \text{clip}\left(\frac{U_i - \overline{U}}{\sigma_U},\, -3,\, 3\right).$$

Let $z_i := (U_i - \overline{U})/\sigma_U$ and $\mathcal{C} = \{i : |z_i| > 3\}$. By centring, $\sum_{i=1}^{G} z_i = 0$, so

$$\sum_{i=1}^{G} \tilde{U}_i \;=\; \sum_{i \notin \mathcal{C}} z_i \;+\; \sum_{i \in \mathcal{C}} \text{sign}(z_i) \cdot 3 \;=\; -\sum_{i \in \mathcal{C}} z_i \;+\; \sum_{i \in \mathcal{C}} \text{sign}(z_i) \cdot 3 \;=\; \sum_{i \in \mathcal{C}} \bigl(\text{sign}(z_i) \cdot 3 - z_i\bigr).$$

For $i \in \mathcal{C}$ we have $|z_i| > 3$, and a direct case split on $\text{sign}(z_i)$ gives $|\text{sign}(z_i) \cdot 3 - z_i| = |z_i| - 3$, so by the triangle inequality

$$\Bigl|\sum_{i=1}^{G} \tilde{U}_i\Bigr| \;\le\; \sum_{i \in \mathcal{C}} \bigl(|z_i| - 3\bigr).$$

By Cauchy–Schwarz, $\sum_{i \in \mathcal{C}} |z_i| \le \sqrt{|\mathcal{C}| \sum_i z_i^2} = \sqrt{|\mathcal{C}|(G-1)}$ (using σ_U unbiased, so $\sum_i z_i^2 = G - 1$); a looser but more interpretable bound uses $|z_i| \le \sqrt{G-1}$ to give $\sum_{i \in \mathcal{C}}\bigl(|z_i| - 3\bigr) \le |\mathcal{C}|(\sqrt{G-1} - 3)$. When $\mathcal{C} = \varnothing$ both sides are exactly 0; note in particular that $|z_i| \le \sqrt{G-1}$ forces $\mathcal{C} = \varnothing$ whenever $G \le 10$, so the looser bound is non-vacuous only in the regime $G \ge 11$ (where $\sqrt{G-1} - 3 > 0$). Substituting into Eq. (6) and using that neither $\gamma_{\text{eff}}(s)$ nor $\overline{|A|}$ depends on i,

$$\Bigl|\sum_{i=1}^{G} A_i^{\text{shaped}} - \sum_{i=1}^{G} A_i\Bigr| \;=\; \gamma_{\text{eff}}(s)\, \overline{|A|}\, \Bigl|\sum_{i=1}^{G} \tilde{U}_i\Bigr| \;\le\; \gamma_{\text{eff}}(s)\, \overline{|A|}\, |\mathcal{C}|\bigl(\sqrt{G-1} - 3\bigr),$$

which is the statement of the proposition.

We emphasise that this is strictly weaker than unbiasedness of the policy-gradient estimator $\sum_i A_i \nabla \log \pi(o_i)$ in the standard sense [Thrampoulidis et al., 2025]: because $\tilde{U}_i$ depends on rollout i through U_i, the substitution $A_i \mapsto A_i^{\text{shaped}}$ does not preserve the expected gradient. The bonus is deliberately biased; the directional bias toward high-MI rollouts is the exploration mechanism. Proposition 1 isolates the bias to the gradient direction by bounding the residual perturbation to the aggregate within-group magnitude. □

9.2 Proof of Proposition 2 (the global maximum is subspace-orthogonal)

Fix a layer ℓ and write $A_k := A_\ell^{(k)} \in \mathbb{R}^{r \times d_{\text{in}}}$. The problem is

$$\max_{A_1,\ldots,A_K} \bigl\|W\bigr\|_*, \qquad W = \bigl[A_1; \ldots; A_K\bigr] \in \mathbb{R}^{Kr \times d_{\text{in}}}, \qquad \|A_k\|_F = c\;\forall k.$$

Let $\sigma_1, \ldots, \sigma_m$ denote the nonzero singular values of W, where $m = \mathrm{rank}(W) \le \min(Kr, d_{\mathrm{in}}) = Kr$ (using the hypothesis $Kr \le d_{\mathrm{in}}$). By Cauchy–Schwarz applied to $(\sigma_1, \ldots, \sigma_m)$ and the all-ones vector,

$$\|W\|_*^2 \;=\; \Big(\sum_{i=1}^{m} \sigma_i\Big)^2 \;\le\; m \sum_{i=1}^{m} \sigma_i^2 \;\le\; Kr \cdot \|W\|_F^2 \;=\; Kr \sum_{k=1}^{K} c^2 \;=\; K^2 r\, c^2,$$

with equality in both inequalities iff $m = Kr$ (i.e. W has full row rank, which requires and is feasible exactly when $Kr \le d_{\mathrm{in}}$, the standing hypothesis) and all Kr singular values are equal. Let s denote this common singular value; then $\|W\|_F^2 = Kr\, s^2 = Kc^2$ gives $s = c/\sqrt{r}$. Under the full-row-rank hypothesis just established, equal singular values are equivalent to $WW^\top = s^2 I_{Kr}$, i.e. the rows of W are mutually orthogonal with common norm s. Restricted to a single block A_k, this forces A_k to have r equal singular values all equal to $s = c/\sqrt{r}$. Across blocks, mutual row orthogonality of W forces the row spaces of $A_1, \ldots, A_K$ to be mutually orthogonal subspaces of $\mathbb{R}^{d_{\mathrm{in}}}$; this is feasible iff $Kr \le d_{\mathrm{in}}$, which holds by hypothesis. The global maximum of $\|W\|_*$ on the constraint set therefore satisfies both conditions, as claimed. □

9.3 Proof of Proposition 3 (TTT-DISCOVER as a special case)

Under either branch the ensemble's mixture $\bar{p}_t$ in Eq. (2) coincides with each member's distribution: in (a) trivially because $K = 1$, and in (b) because identical weights yield identical p_{θ_k}. Each term of the decomposition $H(\bar{p}_t) - \frac{1}{K}\sum_k H(p_{\theta_k})$ then equals the same quantity, so $\mathrm{MI}_t \equiv 0$. Consequently $U_i \equiv 0$, $\overline{U} = 0$, $\sigma_U = 0$, and by the convention stated below Eq. (4) we set $\tilde{U}_i \equiv 0$. The shaped advantage reduces to $A_i^{\mathrm{shaped}} = A_i$ and Eq. (13) reduces to Eq. (12). The hypothesis $\lambda_{\mathrm{NNM}} = 0$ removes $\mathcal{L}_{\mathrm{NNM}}$ from Eq. (8) outright (this assumption is required even in branch (a), since for $K = 1$ the regulariser $-\|A_\ell^{(1)}\|_*$ is a non-trivial function of θ_1 that would otherwise perturb the policy update). What remains of Eq. (8) is $\mathcal{L}_{\mathrm{PG}}(\theta_1) + \lambda_{\mathrm{KL}}\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{base}})$, which is the TTT-DISCOVER objective of Yuksekgonul et al. [2026b]. In branch (b), identical-init plus shared-data training (Sec. 3.5) keeps the K adapters tied at every step, so the per-adapter update is byte-identical to baseline's single-adapter update. □

10 Implementation notes

Chunked MI and base-logprob scoring. Two distinct logit tensors arise during scoring. (i) The *ensemble MI* path materialises an (K, chunk, V) fp32 logit tensor inside the chunked LM head; for Qwen3-8B with $K = 5$, $V = 152{,}064$, and $\mathrm{chunk} = 256$ this peaks at ≈ 0.78 GB per chunk. The MI decomposition of Eq. (2) is summed into a running per-rollout buffer that reuses the same chunk's $\mathrm{logsoftmax}$ (already computed for target-token gather), so the MI contribution costs $< 0.1\%$ of the LM-head matmul. (ii) The *base-model logprob* path needed for the per-token KL-to-base correction (App. 7) materialises a $(1, T, V)$ fp32 logit tensor; for $T = 26$k this would peak at ≈ 16 GB without chunking, and we reduce it to ≈ 8.1 GB by streaming $\mathrm{logsoftmax}$ over the same $\mathrm{chunk} = 256$ granularity and releasing each chunk's allocation before the next is loaded.

Top-7% threshold. Wang et al. [2025] shows that the RL signal in LLMs should concentrate on the high-entropy minority of tokens, the pivotal decision points, rather than be diluted across boilerplate, and AdaDec authors [2025] measure that $\approx 6.75\%$ of decoding steps in code generation are genuinely uncertain with per-task range $[0.85\%, 12.17\%]$. We adopt 7% as a heuristic order-of-magnitude choice; it is not derived from these measurements, which are taken from supervised decoding rather than RL ensemble-MI, but is consistent with them, and it adapts to sequence length so an n-token response contributes $\lceil 0.07n \rceil$ positions, which remains well-calibrated across the 7k–15k-token range typical of reasoning outputs.

Sequential per-adapter backward. A single joint K-adapter forward–backward retains K parallel computation graphs and scales as $\mathcal{O}(K \cdot T \cdot L \cdot D)$ activation memory, exceeding 80 GB on our setup. We therefore iterate over adapters in Stage 5 of Algorithm 1: each adapter's forward and backward are computed in turn, the computation graph is freed before the next adapter's forward, and gradients are accumulated into a shared buffer for the final AdamW step.

Scale-invariant NNM variant. For comparison we also support the Frobenius-normalised variant

$$\widetilde{\mathcal{L}}_{\mathrm{NNM}} \;=\; -\,\frac{1}{L}\sum_{\ell=1}^{L}\frac{\|W_\ell\|_*}{\|W_\ell\|_F}\sqrt{\max(Kr, d_{\mathrm{in}})},$$

which is invariant to the overall scale of the adapter matrices. All experiments in the main paper use the unnormalised form of Eq. (7); the Frobenius-normalised variant behaves similarly in pilots but has not been run at full scale.

Restricting regularisation to $A^{(k)}$. In pilot runs, regularising both $A^{(k)}$ and $B^{(k)}$ produced oscillatory optimisation: a penalty applied to $B^{(k)}$ was absorbed by a compensating rescaling of $A^{(k)}$, and vice versa, giving no net increase in subspace diversity while destabilising the RL loss. This matches the empirical observation of Zhai et al. [2023b]. Leaving $B^{(k)}$ free removes the degeneracy.

11 Streaming MI early-stop and length–reward diagnostic

Mechanism as logged. The streaming early-stop is implemented as a hard cap on the thinking phase: when a rollout reaches 4096 thinking tokens (occasionally 6144 on Erdős where the cap was widened), the windowed ensemble MI is compared to the running 25^{th} percentile of historical MI; if the rollout is below threshold, the thinking phase is closed and the rollout is forced into a bounded code phase. Across all logged streamed rollouts in our four runs the recorded `streaming_mi_stop_step` is the cap itself, so in practice "streamed" means "thinking phase capped at the cap, then a bounded code phase ran." Non-streamed rollouts within the same run either emitted </think> before the cap or were above threshold at the cap and continued to natural termination.

Per-problem firing and savings.

Problem	Firing rate	Mean tokens saved per rollout	Mean tokens saved per firing	Streaming setting in Table 1
AC1	50.0%	569	1,138	enabled
CP26	55.6%	757	1,363	enabled
AC2	disabled	—	—	disabled
Erdős	disabled	—	—	disabled

Paired AC2/Erdős comparison (epochs 3–5). We forked AC2 from a shared epoch-2 checkpoint and ran the remaining three epochs with and without streaming; on Erdős we ran the two configurations as separate full runs and aligned epochs 3–5 for the comparison.

	$R_{\max}$ (stream)	$R_{\max}$ (no stream)	Mean rollout length (stream)	Mean rollout length (no stream)
AC2	0.8415	$\mathbf{0.8563}$	5,312	8,711
Erdős	2.6150	$\mathbf{2.6167}$	5,255	8,720

Length–reward diagnostic. We test whether the within-run rank correlation between rollout length and execution reward separates the four problems. Restricting to correct rollouts and parsing the </think> marker to split thinking from code tokens (streamed rollouts use the logged `phase1_tokens` / `phase2_tokens` fields directly), we report Spearman ρ for epochs 0–2 and for all epochs:

	Epochs 0–2			All epochs		
Problem	ρ_{think}	ρ_{code}	ρ_{total}	ρ_{think}	ρ_{code}	ρ_{total}
AC1	-0.30	$+0.52$	-0.19	-0.51	$+0.48$	-0.43
CP26	$+0.16$	-0.10	$+0.15$	-0.34	-0.14	-0.41
AC2	-0.28	$+0.39$	-0.21	-0.18	$+0.36$	-0.11
Erdős	-0.18	$+0.12$	-0.16	-0.11	$+0.07$	-0.10

The thinking-phase correlation is not significantly positive on any problem and is significantly negative on AC1; the code-phase correlation is positive on AC1 and AC2, weakly positive on Erdős, and null on CP26. No single test on length predicts whether streaming will help or hurt $R_{\max}$ on a given problem; the per-problem calibration in this paper is therefore empirical (paired pilot on AC2/Erdős) rather than diagnosable in advance from a length–reward test.

What streaming-truncated rollouts look like in practice. On AC1 and CP26 the rollouts that achieve the Table 1 $R_{\max}$ are themselves streaming-truncated: AC1 $R_{\max}$=0.6406 from a rollout of length 5,116 (`streaming_mi_stopped`=1); CP26 $R_{\max}$=2.6360 from a rollout of length 5,318 (also truncated). On AC2 and Erdős the Table 1 entries are produced from no-streaming runs (lengths 5,642 and 7,316 respectively). We report this asymmetry directly because it is recoverable from the trajectory logs.

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Algorithm 1 UG-TTT training step (one group of G rollouts).

Require: Ensemble $\{\theta_k\}_{k=1}^{K}$; parent state s from PUCT buffer; group size G.
Require: Coefficients α, β_{ref}, $\gamma_{\max}$, λ_{KL}, λ_{NNM}.
Ensure: Updated ensemble $\{\theta_k\}_{k=1}^{K}$.

// Stage 1: Rollout generation (round-robin over adapters)
Build prompt q from (d, s).
for $i = 1, \dots, G$ **do**
  $k_i \leftarrow \big((i-1) \bmod K\big) + 1$ ▷ 1-indexed; $K \mid G$ so the assignment pattern $(1, 2, \dots, K, 1, 2, \dots)$ repeats identically every group
  $o_i \sim \pi_{\theta_{k_i}}(\cdot \mid q, s)$
  $R_{\mathrm{exec}}^{(i)} \leftarrow \textsc{ProgramChecker}(s, o_i)$
end for
// Stage 2: Mutual-information scoring (single GPU pass)
$\{p_{\theta_k}\}_{k=1}^{K} \leftarrow \textsc{ChunkedScoreAllAdapters}(o_{1:G})$
for $i = 1, \dots, G$ **do**
  for $t = 1, \dots, |o_i|$ **do**
    $\bar{p}_t \leftarrow \frac{1}{K}\sum_{k=1}^{K} p_{\theta_k}(\cdot \mid q, o_{<t}^{(i)})$
    $\mathrm{MI}_t^{(i)} \leftarrow H(\bar{p}_t) - \frac{1}{K}\sum_{k=1}^{K} H\big(p_{\theta_k}(\cdot \mid q, o_{<t}^{(i)})\big)$ ▷ Eq. (2)
  end for
  $U_i \leftarrow \text{top-7\%-mean}_t\, \mathrm{MI}_t^{(i)}$ ▷ Eq. (3)
end for
// Stage 3: Adaptive temperature and LOO advantage (on R_{exec} only)
$\beta(s) \leftarrow \textsc{BisectionSolve}\big\{\beta > 0 : \mathrm{KL}(q_\beta \,\|\, \mathrm{Unif}) = \ln 2\big\}$ on $\{R_{\mathrm{exec}}^{(i)}\}$ ▷ Eq. (10)
for $i = 1, \dots, G$ **do**
  $w_i \leftarrow \exp(\beta R_{\mathrm{exec}}^{(i)}) \big/ \frac{1}{G-1}\sum_{j \neq i} \exp(\beta R_{\mathrm{exec}}^{(j)})$
  $A_i \leftarrow w_i - 1$ ▷ Eq. (11)
end for
// Stage 4: MI-shaped advantage
$\overline{U} \leftarrow \frac{1}{G}\sum_i U_i$; $\sigma_U \leftarrow \mathrm{std}_{\mathrm{unbiased}}(U)$ (cf. Eq. (4)); $\overline{|A|} \leftarrow \frac{1}{G}\sum_j |A_j|$
$\gamma_{\mathrm{eff}}(s) \leftarrow \alpha \cdot \min\big(\beta(s)/\beta_{\mathrm{ref}},\ \gamma_{\max}\big)$ ▷ Eq. (5)
for $i = 1, \dots, G$ **do**
  $\tilde{U}_i \leftarrow \mathrm{clip}\big((U_i - \overline{U})/\sigma_U,\ -3,\ 3\big)$ ▷ Eq. (4)
  $A_i^{\mathrm{shaped}} \leftarrow A_i + \gamma_{\mathrm{eff}}(s) \cdot \overline{|A|} \cdot \tilde{U}_i$ ▷ Eq. (6)
end for
if $\lambda_{\mathrm{KL}} > 0$ **then**
  Broadcast A_i^{shaped} per-token; add token-wise KL correction against π_{base} (Appendix 7)
end if
// Stage 5: Per-adapter backward, NNM backward, then optimiser step
for $k = 1, \dots, K$ **do**
  Compute $\frac{1}{K}\mathcal{L}_{\mathrm{PG}}^{(k)}(\theta_k)$ over all G rollouts and call `backward()`; gradients accumulate into θ_k.grad ▷ Eq. (13)
  Release adapter k's computation graph before advancing
end for
  // KL-to-base contribution is already inside $\mathcal{L}_{\mathrm{PG}}^{(k)}$ via the Stage 4 token-wise correction (App. 7)
Compute $\lambda_{\mathrm{NNM}}\, \mathcal{L}_{\mathrm{NNM}}$ and call `backward()`; gradients accumulate into all $\{\theta_k\}$ ▷ Eq. (7)
for $k = 1, \dots, K$ **do**
  AdamW step on θ_k using its accumulated gradient; zero θ_k.grad
end for